



# IRIS

INTELLIGENT REAL-TIME  
ILLUMINATION SYSTEM



**“OR lights are frequently adjusted,  
notably every 7.5 minutes, and 50%  
of previously sterile light handles  
have been shown to foster  
bacterial growth.”**

<https://pmc.ncbi.nlm.nih.gov/articles/PMC7305019/>

**“Over 40% of surgeons complain about eye fatigue, and fatigue significantly increases the risk of surgical complications.”**

<https://pmc.ncbi.nlm.nih.gov/articles/PMC8302286/>

# Surgeon Interviews



“In general overhead lights are good but our **heads get in the way**. Headlights are then necessary but are often **not bright enough** or add **weight and discomfort to our necks**.”

—Dr. Badgwell (Gastric Surgical Oncologist at M.D. Anderson)



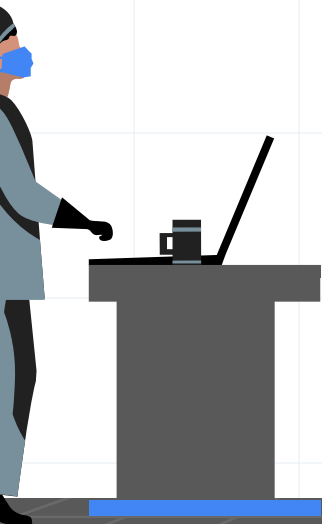
“**Lights are generally suboptimal**. Headlights either **tether you in place** or are subject to **short battery life**, additionally, you have to remember to put them on”

—Dr. Baldwin (Associate Professor of Orthopaedic Surgery at Children's Hospital of Philadelphia)

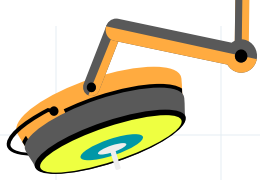


“Light handles are **frequently contaminated** when a head or arm hits them. It is a **major interruption** to replace the sterile light handle cover and adds cost. The lighting is often delivered in a collimated beam that causes constant readjustment of light. There is a need for a **smart light** that **cannot be contaminated**, **doesn't need adjustment**, and **delivers wavelengths that illuminate proper tissue color**.”

—Dr. Sheckter (Associate Professor of Surgery at Stanford University School of Medicine)



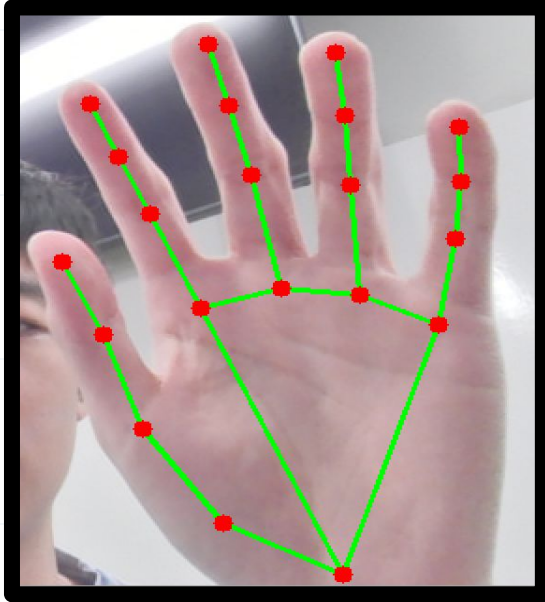
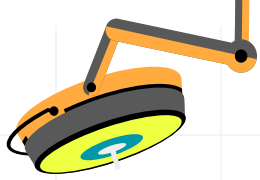
**So why hasn't surgical  
lighting evolved to  
meet the demands of  
modern medicine?**



# Meet IRIS



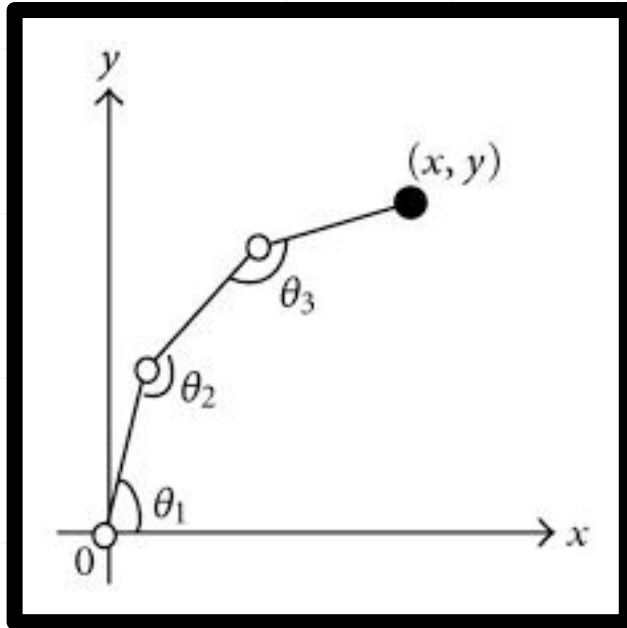
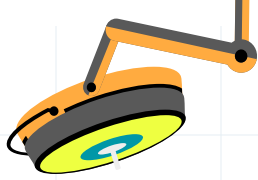
- Tracks the Surgical Field Automatically
- Eliminates Manual Light Adjustments
- Maintains Continuous Surgical Focus
- The **light moves**, not the surgeon.



## How the System Sees

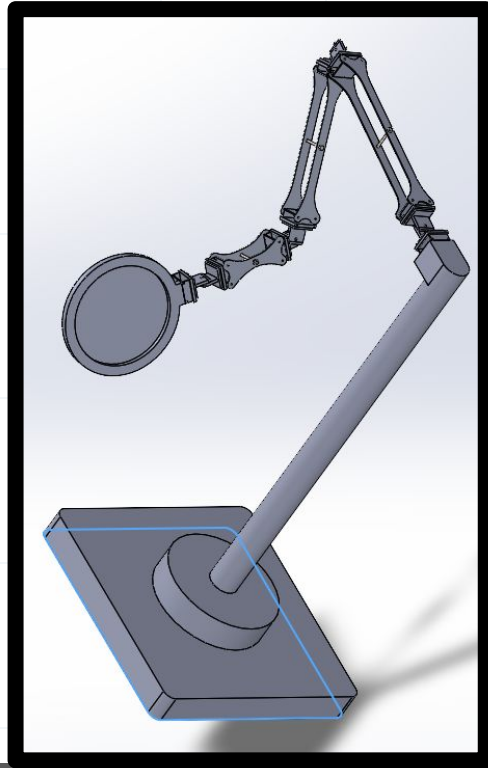
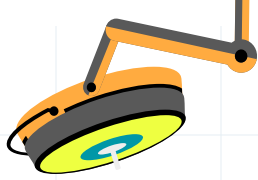
- Captures the surgical field in real time
- Processes video using computer vision
- Tracks the target area continuously
- Calculates optimal light positioning
- Updates dynamically with feedback





## From Vision to Movement

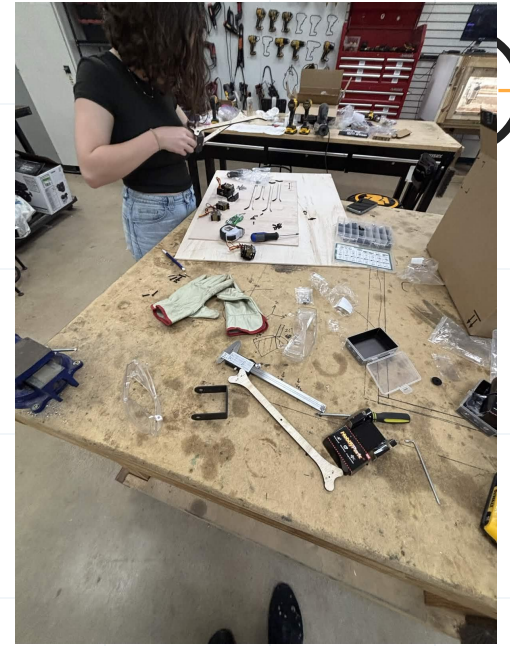
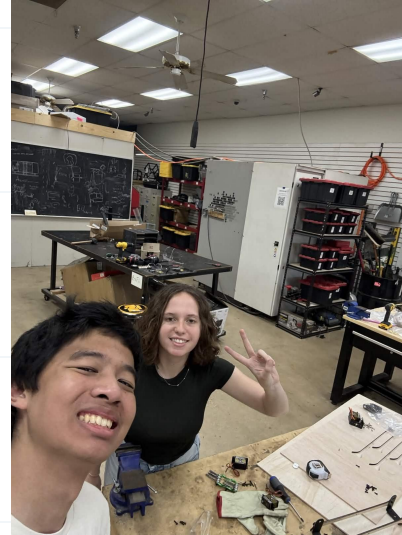
- Detect Target Position  $(x, y)$
- Convert to Real-World Direction
- Map to Motor Angles
- Send Commands to Motors
- Adjust Light Automatically



## The Mechanism

### Our Design:

- 3-DOF Articulation
- Lightweight Acrylic Arms
- Motorized Pivot Joints
- Mobile Stand



# Cost Economics



40,000+

operating rooms in the U.S.

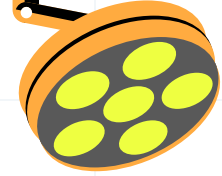
\$13 Billion

in costs due to surgical errors

90 Million

surgeries per year

# Market Opportunity



## Understanding The Numbers

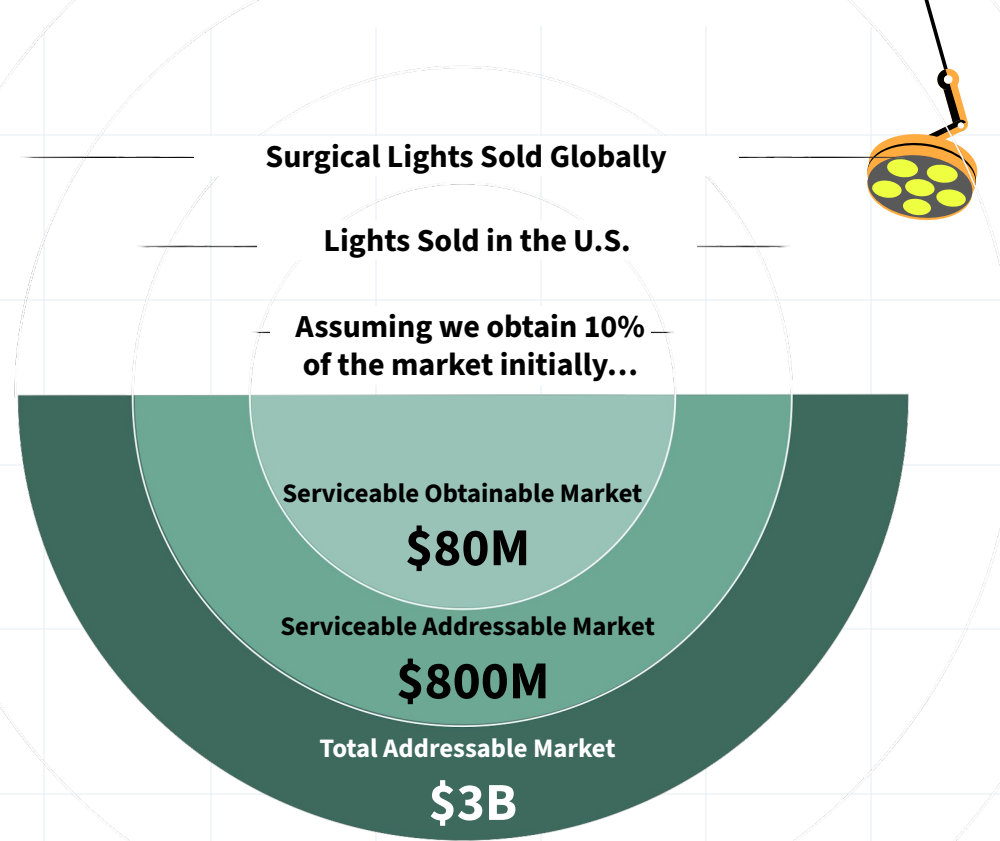
A single hospital does around 5,000 surgeries per year.

Each minute is worth at least \$100 in the OR.

If we can just save 10 minutes for each surgery:

$5,000 \times 10 \times 100 =$

**\$5 million dollars saved per hospital each year.**



# Business Model



## B2B Sales

Sell directly to hospitals or license the technology to Medtech companies.



## Service Contracts

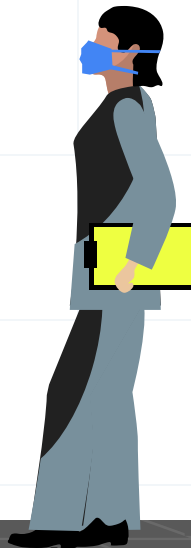
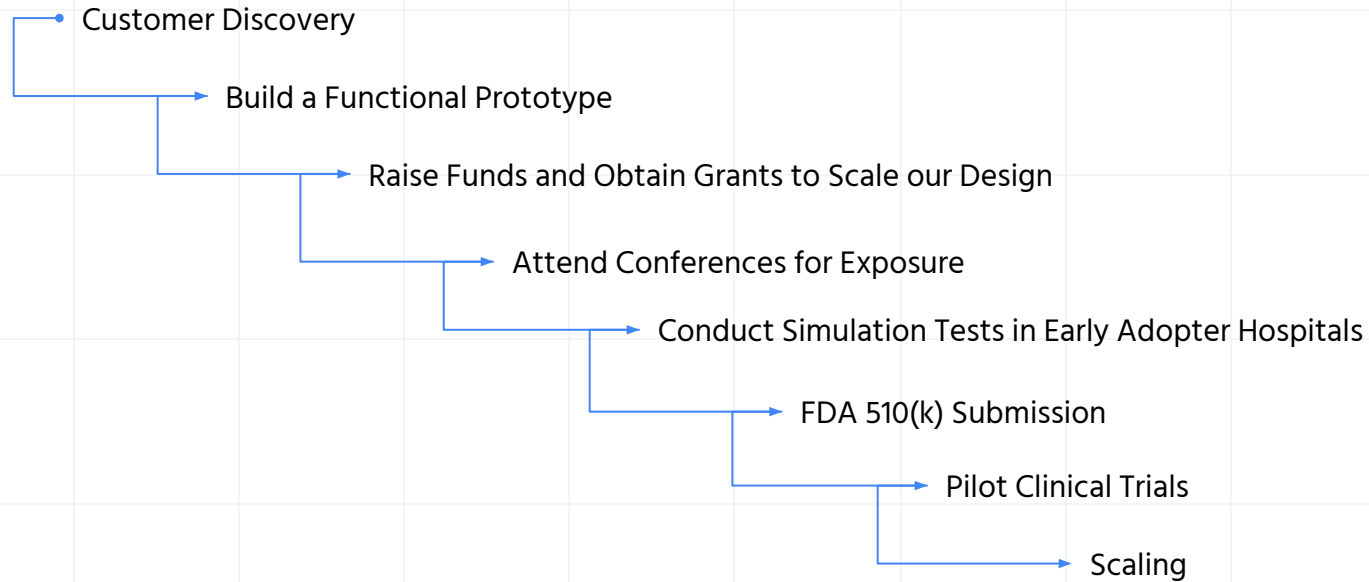
Make contracts with hospitals to provide service and maintenance for the lights sold.



## Cost per Light

We estimate that the light will be sold for ~\$50,000 each, which is on the lower side for more advanced lighting systems.

# Timeline





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